



```
In [2]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
In [3]: # Load the dataset from the correct path
df = pd.read_csv(r"C:\Users\Neethu\Documents\ML\archive\hotel_bookings.csv")
df.head()
```

```
Out[3]:
```

	hotel	is_canceled	lead_time	arrival_date_year	arrival_date_month	arrival_d
0	Resort Hotel	0	342	2015	July	
1	Resort Hotel	0	737	2015	July	
2	Resort Hotel	0	7	2015	July	
3	Resort Hotel	0	13	2015	July	
4	Resort Hotel	0	14	2015	July	

5 rows × 32 columns

```
In [5]: df = df.dropna()

print("Dataset Shape:", df.shape)

column = "adr" # Average Daily Rate

print("\n----- CENTRAL TENDENCY -----")
print("Mean:", df[column].mean())
print("Median:", df[column].median())
print("Mode:", df[column].mode()[0])

plt.figure(figsize=(8,5))
sns.histplot(df[column], bins=30, kde=True)

plt.axvline(df[column].mean(),
            color='red',
            linestyle='--',
            label='Mean')

plt.axvline(df[column].median(),
            color='green',
            linestyle='--',
            label='Median')

plt.title("Central Tendency of ADR")
plt.legend()
```

```

plt.show()

outlier_rows = pd.DataFrame({
    'hotel': ['Resort Hotel']*3,
    'lead_time': [1000, 1200, 1500],
    'arrival_date_year': [2017]*3,
    'arrival_date_month': ['August']*3,
    'arrival_date_week_number': [35]*3,
    'arrival_date_day_of_month': [15]*3,
    'stays_in_weekend_nights': [2]*3,
    'stays_in_week_nights': [5]*3,
    'adults': [2]*3,
    'children': [0]*3,
    'babies': [0]*3,
    'meal': ['BB']*3,
    'country': ['PRT']*3,
    'market_segment': ['Online TA']*3,
    'distribution_channel': ['TA/TO']*3,
    'is_repeated_guest': [0]*3,
    'previous_cancellations': [0]*3,
    'previous_bookings_not_canceled': [0]*3,
    'reserved_room_type': ['A']*3,
    'assigned_room_type': ['A']*3,
    'booking_changes': [0]*3,
    'deposit_type': ['No Deposit']*3,
    'agent': [1]*3,
    'company': [1]*3,
    'days_in_waiting_list': [0]*3,
    'customer_type': ['Transient']*3,
    'adr': [1000, 1500, 2000],
    'required_car_parking_spaces': [0]*3,
    'total_of_special_requests': [0]*3,
    'reservation_status': ['Check-Out']*3,
    'reservation_status_date': ['2017-08-15']*3
})

df = pd.concat([df, outlier_rows], ignore_index=True)

Q1 = df['adr'].quantile(0.25)
Q3 = df['adr'].quantile(0.75)

IQR = Q3 - Q1

lower = Q1 - 1.5 * IQR
upper = Q3 + 1.5 * IQR

outliers = df[(df['adr'] < lower) | (df['adr'] > upper)]

print("\n----- OUTLIERS -----")
print("Number of Outliers:", len(outliers))
print(outliers[['adr']].head())

```

```

plt.figure(figsize=(8,5))
sns.boxplot(y=df['adr'])
plt.title("Outlier Detection using Boxplot")
plt.show()

plt.figure(figsize=(8,5))
sns.scatterplot(
    x='lead_time',
    y='adr',
    data=df
)

plt.title("Lead Time vs ADR")
plt.xlabel("Lead Time")
plt.ylabel("ADR")
plt.show()

correlation = df['lead_time'].corr(df['adr'])

print("\nCorrelation:")
print(correlation)

plt.figure(figsize=(8,5))
sns.boxplot(
    x='hotel',
    y='adr',
    data=df
)

plt.title("Hotel Type vs ADR")
plt.show()

print("\nAverage ADR by Hotel Type")
print(df.groupby('hotel')['adr'].mean())

cross_tab = pd.crosstab(
    df['hotel'],
    df['customer_type']
)

print("\nCross Tabulation")
print(cross_tab)

cross_tab.plot(
    kind='bar',
    figsize=(10,6)
)

plt.title("Hotel Type vs Customer Type")
plt.ylabel("Count")
plt.show()

```

```
plt.figure(figsize=(8,5))
sns.heatmap(
    cross_tab,
    annot=True,
    cmap='YlGnBu'
)

plt.title("Hotel Type vs Customer Type Heatmap")
plt.show()
```

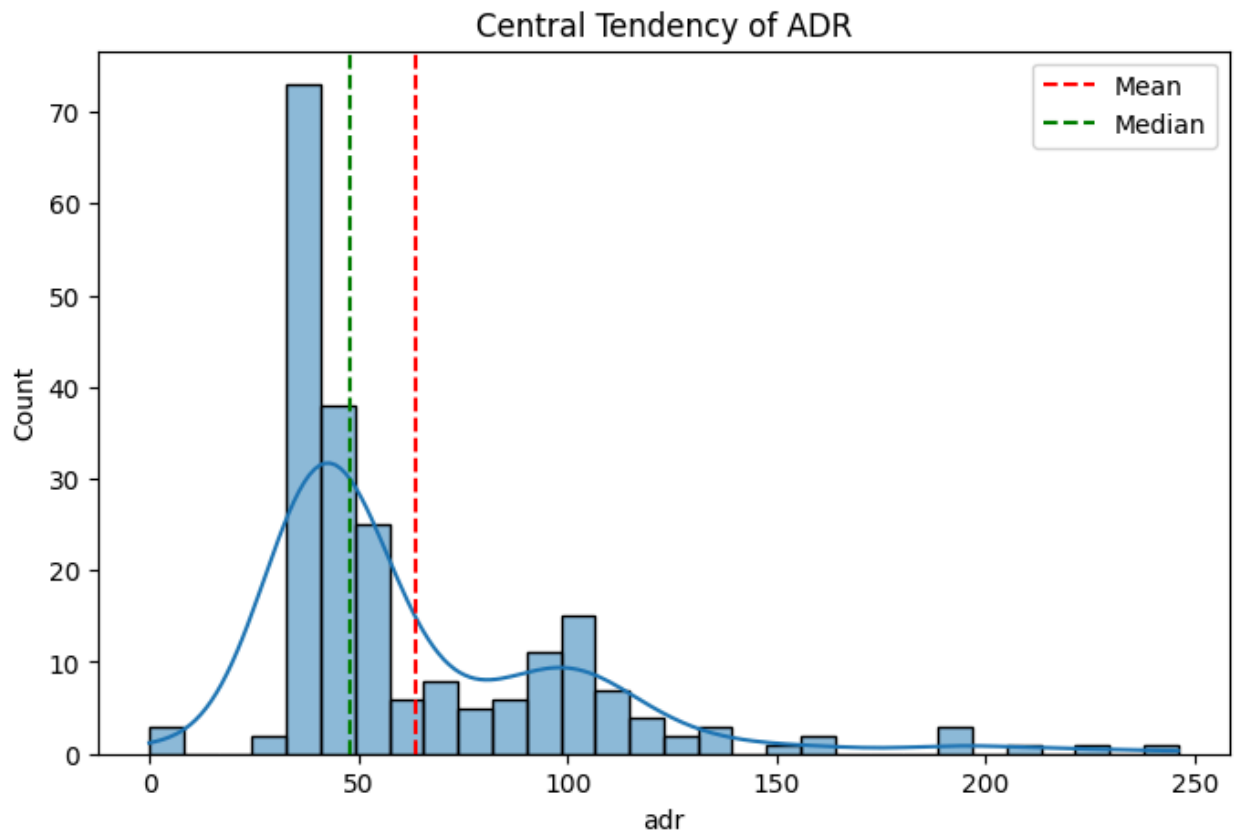
Dataset Shape: (217, 32)

----- CENTRAL TENDENCY -----

Mean: 63.79373271889401

Median: 48.0

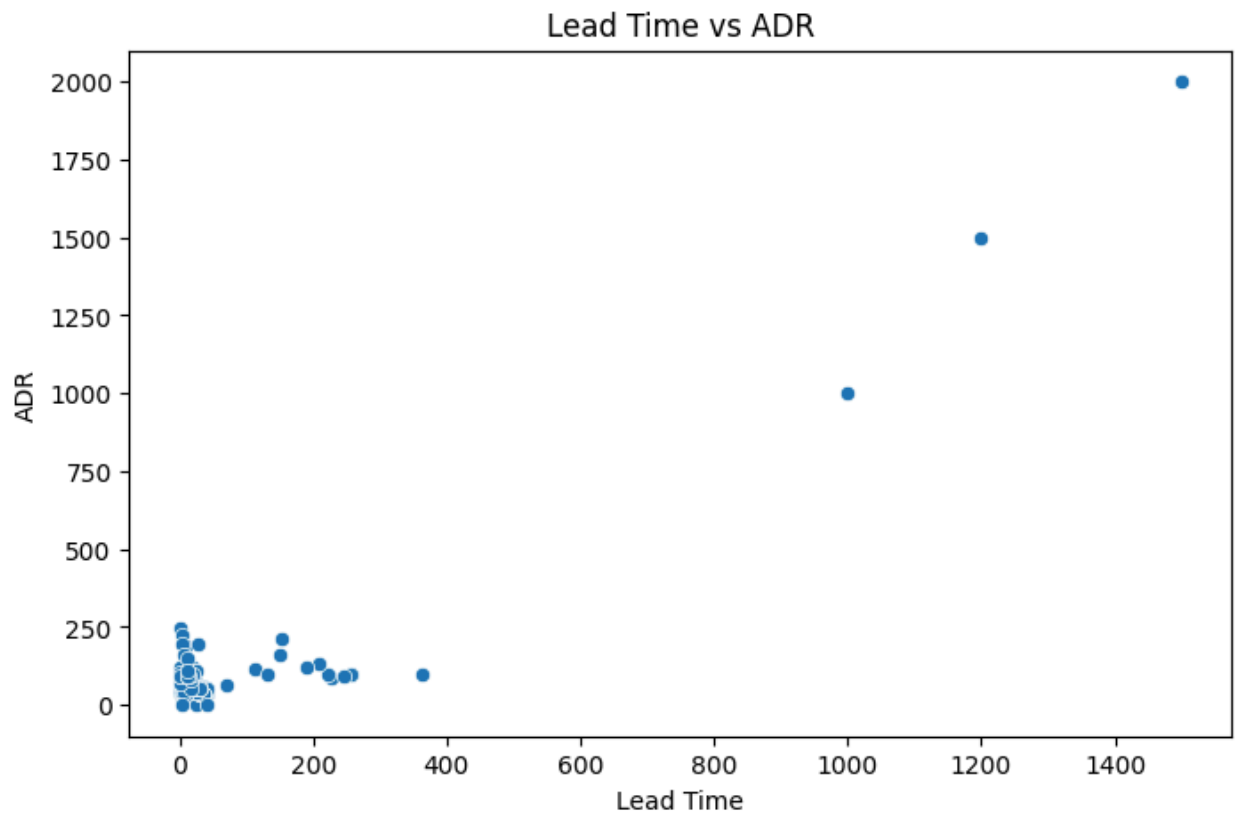
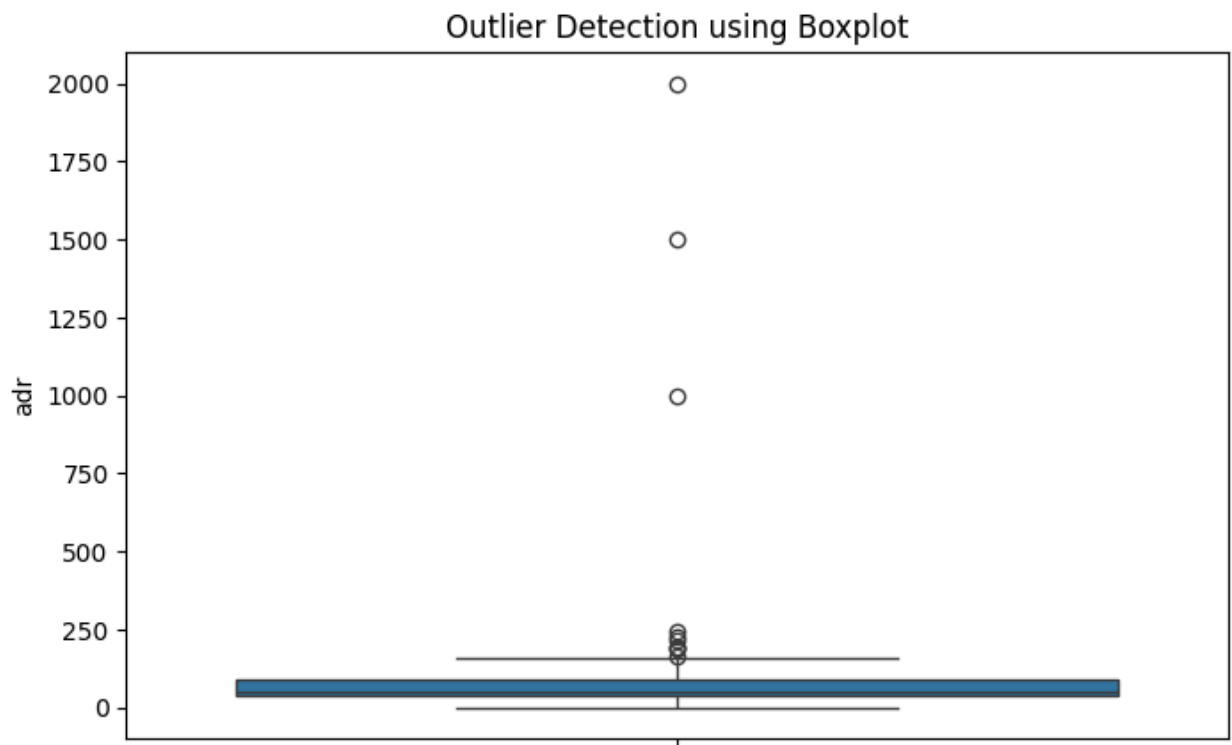
Mode: 36.0



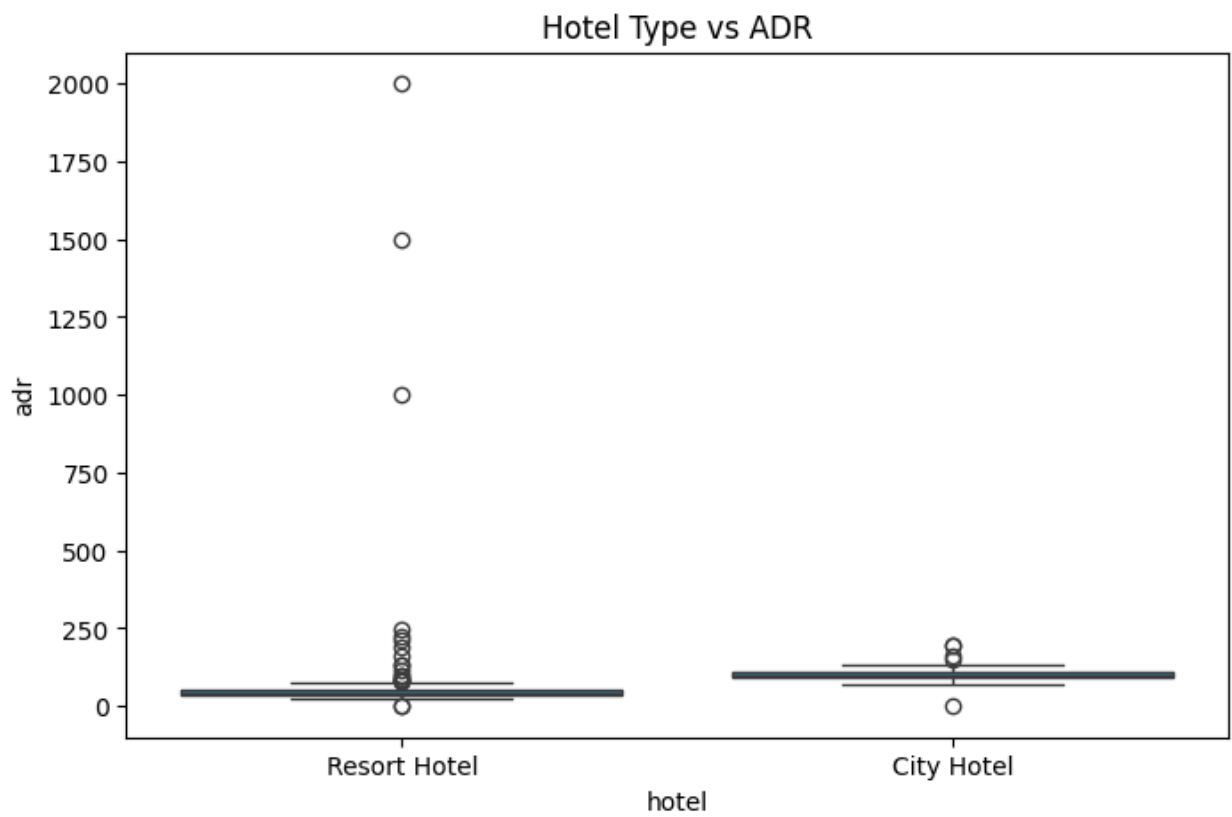
----- OUTLIERS -----

Number of Outliers: 10

	adr
151	189.0
152	246.0
153	224.0
154	212.6
167	163.4



Correlation:
0.9103335829535049



Average ADR by Hotel Type

```

hotel
City Hotel      106.207447
Resort Hotel    77.176243
Name: adr, dtype: float64

```

Cross Tabulation

```

customer_type  Contract  Group  Transient  Transient-Party
hotel
City Hotel      2        2        30         13
Resort Hotel    3        6        30        134

```

